Student Academic Performance

Analysis Report & Insights

Target Variable: Cumulative GPA (last semester / 4.00)

Dataset: Students.csv | April 2026

BY MONYAI RITSHIDZE ZENYXANALYTICS

1. Overview of the Dataset

This analysis investigates factors that influence student GPA (measured on a 4.00 scale) at a university in Cyprus. The dataset (Students.csv) contains responses from students across a wide range of variables, categorised below. Data cleaning was performed by removing rows with missing values to ensure analytical integrity.

1.1 Variable Categories

The dataset contains 25 variables spanning three thematic domains:

Category	Number of Variables	Example Variables
Demographic & Family Background	10	Scholarship type, Do you have a partner?
Academic Engagement	12	Weekly study hours, Reading frequency
Lifestyle & Logistics	3	Additional work, Sports activity

2. Methodology

The Python script (analyse.py) applies a three-stage analytical pipeline to each variable in relation to the target GPA:

Stage	Method	Purpose
1. Visualisation	Scatter plots (matplotlib)	Explore the visual relationship between each variable and GPA
2. Descriptive Statistics	Mean, Median, Std Dev, Variance, CV, Percentiles (pandas/numpy)	Summarise the distribution of each variable
3. Correlation & Regression	Pearson Correlation + Simple Linear Regression (scipy.stats)	Quantify the strength and direction of each variable's impact on GPA
4. Heatmap	Full correlation matrix (seaborn)	Identify inter-variable relationships across the entire dataset

3. Descriptive Statistics — Key Insights

Descriptive statistics are computed for every variable. Below are the statistical measures calculated and what they reveal about the student population:

Statistic	What It Tells Us
Mean	The average value across all students — the central tendency of each variable
Median	The middle value; more robust than mean when outliers are present
Standard Deviation	How spread out student responses are — high SD means high variability
Variance	Square of SD; shows overall dispersion in responses
Coefficient of Variation (%)	Relative variability — useful for comparing spread across variables with different scales
25th / 50th / 75th Percentile	Shows distribution shape; the interquartile range (Q3-Q1) indicates concentration of data

3.1 Interpretation Guide

- A low CV (< 30%) indicates students tend to respond similarly for that variable (e.g., attendance may be consistently high).

- A high CV (> 50%) suggests diverse student behaviour, making the variable potentially more discriminating for GPA prediction.
- Variables where mean and median diverge significantly suggest skewed distributions — a few outlier students are pulling the average.

4. Correlation Analysis

Pearson correlation coefficients are computed between each variable and GPA. The script applies a filter: only variables with standard deviation < 0.5 AND coefficient of variation < 0.5 are reported. This targets variables that are relatively stable (low noise) yet still meaningfully correlated.

4.1 Correlation Interpretation Scale

Correlation Range	Strength	Implication for GPA
0.70 to 1.00	Strong Positive	As this variable increases, GPA reliably increases — key predictor
0.40 to 0.69	Moderate Positive	Meaningful positive relationship — worth including in models
0.10 to 0.39	Weak Positive	Slight tendency — may not be reliable on its own
0.00 to 0.09	Negligible	Little to no linear relationship with GPA
-0.10 to -0.39	Weak Negative	Slight inverse tendency — higher values slightly linked to lower GPA
-0.40 to -1.00	Moderate/Strong Negative	Increased values are associated with lower GPA — potential risk factor

4.2 Variables Most Likely to Correlate with GPA

Based on the variable categories identified in the dataset, the following groups are most likely to yield meaningful correlations:

- Academic engagement variables (weekly study hours, attendance, exam preparation) — typically strong positive correlates of GPA in student performance literature.
- Note-taking and active listening in class — associated with deeper learning and typically positively correlated with academic outcomes.
- Seminar/conference attendance — indicator of academic motivation, likely positively associated.
- Flip-classroom participation — innovative pedagogy; effect direction depends on student adaptation.

- Additional work (employment) — may negatively affect GPA due to reduced study time if hours are high.
- Scholarship type — may correlate as scholarship holders often maintain minimum GPA requirements.

5. Linear Regression Analysis

Simple linear regression is applied for every variable against GPA using scipy.stats.linregress. Each model produces the following outputs:

Output	Symbol	Meaning
Gradient (Slope)	m	How much GPA changes for each 1-unit increase in the variable
Intercept	b	Predicted GPA when the variable equals zero
R-value	r	Pearson correlation coefficient between variable and GPA
P-value	p	Statistical significance — $p < 0.05$ means the relationship is unlikely due to chance
Standard Error	SE	Precision of the slope estimate — lower is more reliable

5.1 How to Use Regression Results

- Gradient > 0: Positive predictor — each unit increase in the variable predicts higher GPA.
- Gradient < 0: Negative predictor — each unit increase predicts lower GPA (potential risk factor).
- Check p-value first: A low p-value (< 0.05) confirms the gradient is statistically meaningful, not random noise.
- R-squared (r^2): Square the r-value to get the proportion of GPA variance explained by the variable alone.
- A variable with $r = 0.50$ explains 25% of GPA variance — meaningful but not the whole picture.

6. Correlation Heatmap

The seaborn heatmap visualises the full correlation matrix across all numeric variables. This reveals not just variable-to-GPA relationships but also inter-variable relationships. Key patterns to look for:

- Deep red clusters (correlation near +1.00): Variables that strongly move together — potential multicollinearity if used in multiple regression.
- Deep blue clusters (correlation near -1.00): Variables with strong inverse relationships.
- GPA column/row: The column or row for the target variable reveals at a glance which predictors are strongest.
- Diagonal: Always 1.00 (a variable is perfectly correlated with itself) — confirms data integrity.
- Symmetric pattern: The matrix is symmetric about the diagonal — each pair appears twice.

7. Analytical Recommendations

Based on the methodology and variable structure, the following enhancements would strengthen the analysis:

7.1 Script Improvements

- Relax correlation filter: The current condition (std < 0.5 AND CV < 0.5) may exclude many important variables. Consider reporting all correlations and filtering only by p-value significance.
- Handle NaN in regression: linregress fails on NaN values — add dropna() before each regression call to prevent errors.
- Multiple regression: After identifying top individual predictors, combine them into a multivariate OLS model (sklearn or statsmodels) for a more realistic predictive model.
- Categorical encoding: Ordinal variables (scholarship type, parental status) should be explicitly mapped to numeric codes before analysis.
- Outlier detection: Add IQR-based outlier removal after the current dropna() step for cleaner statistical outputs.

7.2 Expected High-Impact Variables

Based on published student performance research, these dataset variables are most likely to show meaningful GPA correlations:

Variable	Expected Direction	Likely Strength
Weekly study hours	Positive	Strong
Attendance in class	Positive	Strong
Preparation to midterm exams	Positive	Strong
Taking notes in classes	Positive	Moderate
Listening in classes	Positive	Moderate

Variable	Expected Direction	Likely Strength
Discussion improves interest	Positive	Moderate
Additional work (employment)	Negative	Moderate
Sports activity	Mixed	Weak–Moderate
Number of siblings	Negligible	Weak
Transportation to university	Negative	Weak

8. Summary

This analysis provides a comprehensive multi-method examination of student GPA determinants using a 25-variable dataset. The pipeline progresses logically from visualisation through descriptive statistics, correlation, and predictive modelling. Key takeaways:

- Academic behaviour variables (study habits, attendance, preparation) are expected to be the strongest GPA predictors.
- Demographic and family variables provide contextual richness but weaker direct GPA effects.
- The linear regression models provide per-variable prediction equations that quantify the academic return on each behaviour.
- The correlation heatmap enables holistic insight into the student environment as a connected system.
- Refining the script's filtering logic and adding multivariate modelling would significantly increase analytical power.

Compiled from analyse.py and Students.csv dataset | April 2026